

Shiv Nadar University Chennai

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
Subject Code & Name	CS3801 – Web Technologies Lab	AY: 2026–27

Assignment - 9 Student Information Web Server

1 Aim:

To Develop a Student Information Web Server Using Node.js Develop a Student Information Web Server using Node.js that demonstrates the use of the built-in File System (fs), HTTP (http), and URL (url) modules.

2 Code Implementation:

2.1 server.js

```
JS server.js > http.createServer() callback
1  const http = require("http");
2  const fs = require("fs");
3  const url = require("url");
4
5  const PORT = 3000;
6
7  const server = http.createServer((req, res) => {
8
9      // Parse the requested URL
10     const parsedUrl = new URL(
11         req.url,
12         `http://${req.headers.host}`
13     );
14
15     const pathname = parsedUrl.pathname;
16     const query = parsedUrl.searchParams;
17     // Home page
18     if (pathname === "/" ) {
19
20         res.writeHead(200, {
21             "Content-Type": "text/html"
22         });
23
24         res.end(`
25             <h1>Student Information Web Server</h1>
26
27             <h2>Available Operations</h2>
28
29             <ul>
30                 <li><a href="/students">View All Students</a></li>
31                 <li><a href="/student?id=101">View Student 101</a></li>
32             </ul>
33         `);
34     }
35
36     // Display all students
37     else if (pathname === "/students") {
38
39         fs.readFile("students.json", "utf8", (err, data) => {
40
41             if (err) {
42
43                 res.writeHead(500, {
44                     "Content-Type": "text/plain"
45                 });
46
47                 res.end("Error reading student data");
48                 return;
49             }
50
51             const students = JSON.parse(data);
52
53             res.writeHead(200, {
54                 "Content-Type": "text/plain"
55             });
56
57             res.end(JSON.stringify(students));
58         });
59     }
60
61     // Default response for unknown paths
62     else {
63         res.writeHead(404, {
64             "Content-Type": "text/plain"
65         });
66         res.end("404 Not Found");
67     }
68
69 });
70
71 server.listen(PORT, () => {
72     console.log(`Server is running on port ${PORT}`);
73 });
```

```

53     res.writeHead(200, {
54       "Content-Type": "text/html"
55     });
56
57     let html = `
58       <h1>Student Information</h1>
59       <table border="1" cellpadding="10">
60         <tr>
61           <th>ID</th>
62           <th>Name</th>
63           <th>Department</th>
64           <th>Year</th>
65           <th>CGPA</th>
66         </tr>
67       `;
68
69       students.forEach(student => {
70
71         html += `
72           <tr>
73             <td>${student.id}</td>
74             <td>${student.name}</td>
75             <td>${student.department}</td>
76             <td>${student.year}</td>
77             <td>${student.cgpa}</td>
78           </tr>
79         `;
80       });
81
82       html += "</table>";
83
84       res.end(html);
85     });
86   }
87
88   // Display a particular student
89   else if (pathname === "/student") {
90
91     const studentId = query.id;
92
93     fs.readFile("students.json", "utf8", (err, data) => {
94
95       if (err) {
96
97         res.writeHead(500, {
98           "Content-Type": "text/plain"
99         });
100
101         res.end("Error reading student data");
102         return;

```

```

102         return;
103     }
104
105     const students = JSON.parse(data);
106
107     const student = students.find(
108         s => s.id == studentId
109     );
110
111     res.writeHead(200, {
112         "Content-Type": "text/html"
113     });
114
115     if (student) {
116
117         res.end(`
118             <h1>Student Details</h1>
119
120             <p><strong>ID:</strong> ${student.id}</p>
121             <p><strong>Name:</strong> ${student.name}</p>
122             <p><strong>Department:</strong> ${student.department}</p>
123             <p><strong>Year:</strong> ${student.year}</p>
124             <p><strong>CGPA:</strong> ${student.cgpa}</p>
125         `);
126
127     } else {
128
129         res.end(`
130             <h1>Student Not Found</h1>
131             <p>No student exists with ID ${studentId}</p>
132         `);
133     }
134 });
135 }
136
137 // Invalid URL
138 else {
139
140     res.writeHead(404, {
141         "Content-Type": "text/html"
142     });
143
144     res.end(`
145         <h1>404 - Page Not Found</h1>
146     `);
147 }
148
149 });
150
151 server.listen(PORT, () => {
152
153     console.log(
154         `Student Information Server running at http://localhost:\${PORT}`
155     );
156
157 });

```

2.2 Students.js

```
() students.json > ...
1  [
2    {
3      "id": 101,
4      "name": "Rahul",
5      "department": "CSE",
6      "year": 2,
7      "cgpa": 8.5
8    },
9    {
10     "id": 102,
11     "name": "Priya",
12     "department": "AIDS",
13     "year": 2,
14     "cgpa": 9.1
15   },
16   {
17     "id": 103,
18     "name": "Arun",
19     "department": "ECE",
20     "year": 3,
21     "cgpa": 8.2
22   },
23   {
24     "id": 104,
25     "name": "Sneha",
26     "department": "CSE",
27     "year": 1,
28     "cgpa": 9.0
29   }
30 ]
```

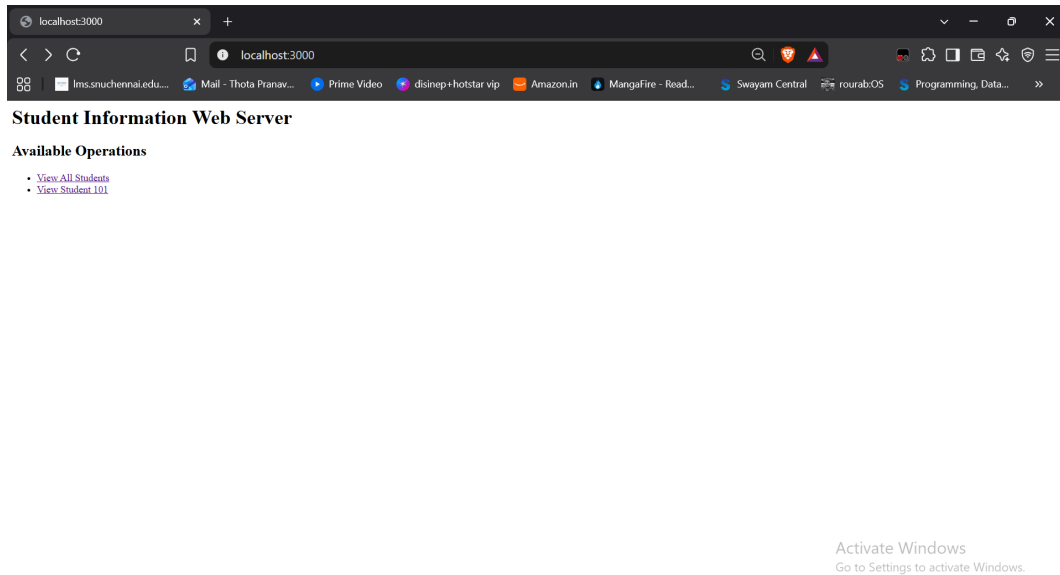
Problems Output Debug Console **Terminal** Ports

* History restored

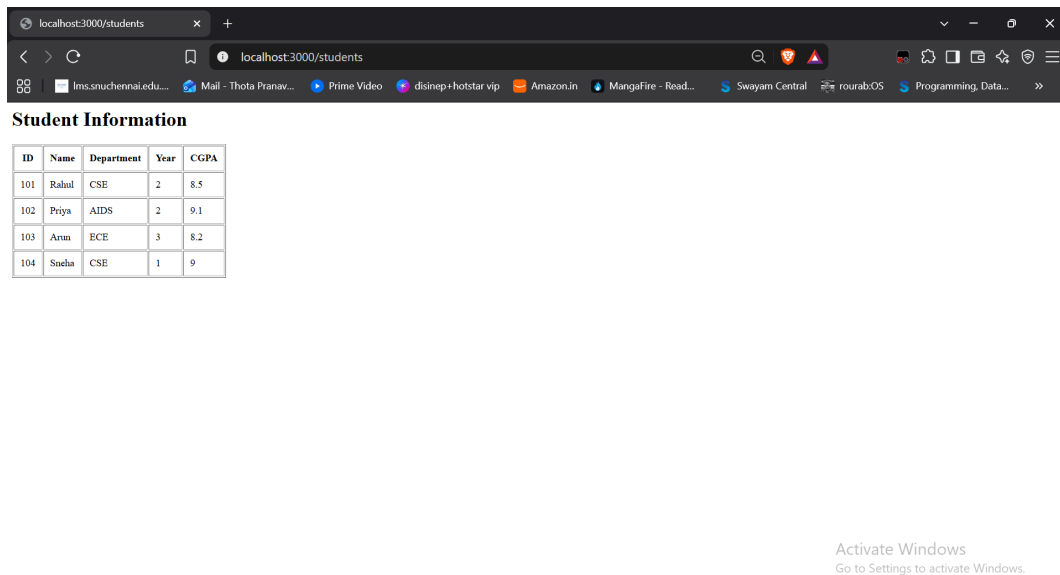
PS C:\Users\lenova\Downloads\SEM 5\Web Technologies\Lab\Lab-9\Student_Server> node server.js
Student Information Server running at http://localhost:3000
█

3 Web Page:

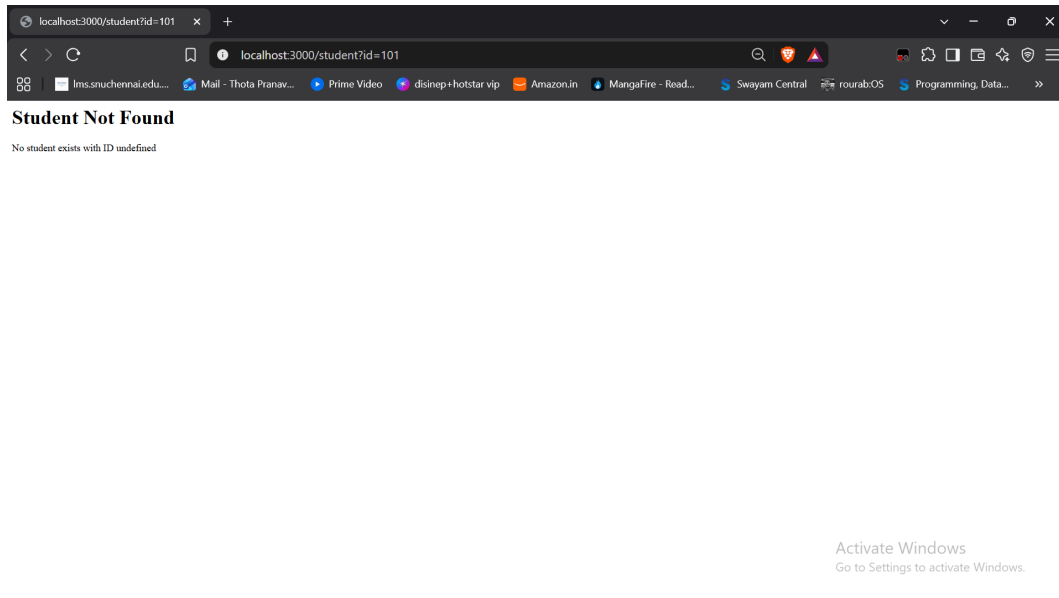
3.1 Home:



3.2 Students Data



3.3 Find student 101



4 Conclusion

The experiment successfully demonstrated the development of a Student Information Web Server using Node.js. The `http` module was used to create the web server, the `fs` module was used to read student data from a JSON file, and the `url` module was used to parse URLs and query parameters. Thus, the experiment demonstrated how Node.js built-in modules can be combined to create a simple and functional web server for managing student information.

5 Code Link:

<https://github.com/PranavVarshin/Web-Technologies-Lab/tree/main/Lab-9>

- NAME: T Pranav Varshin(24110413)
- Reg No: 24011101116
- Section: AIDS-B